Huawei Access Node JAY-ACC-35 Routing Protocol Flap Resolution SOP

Node: JAY-ACC-35 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_7152

Description

In the context of the Huawei Access Node JAY-ACC-35 at the Jayanagar Metro Site, a Routing Protocol Flap Failure, as indicated by Alarm Code ALARM_7152 of Major severity, can occur due to excessive BGP route churn, OSPF adjacency loss, or ISIS LSP flooding resulting from misconfigured or malfunctioning routing protocols, potentially triggered by hardware or software malfunctions, or even malicious activities. This failure domain is specific to the Access role on Huawei equipment due to its reliance on precise routing protocol configuration and optimization. Inadequate or inconsistent configuration can lead to routing instability, causing frequent route flaps, which in turn can result in network congestion, packet loss, and ultimately, service outages. The underlying protocol mechanics involve the dynamic exchange of routing information between nodes, with BGP, OSPF, and ISIS playing critical roles in establishing and maintaining routing table consistency. Hardware mechanics, such as faulty interfaces or power supply issues, can also contribute to this failure domain by disrupting communication between nodes, thereby affecting routing protocol operations.

Pre-requisite Checks

- Verify that the Jayanagar Metro Site is accessible and that the dispatching team has been notified.

- Check the status of redundant links to ensure they are operational and can take over in case of a failure.

- Confirm that all necessary permissions and access rights have been granted to the maintenance personnel.

Potential Root Causes

Cause	Probability
Misconfigured or inconsistent routing protocol settings	High
Hardware or software malfunction	Medium
Malicious activities or unauthorized changes	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table ip-prefix 0.0.0.0/0, Expected Output: Routing table for VPN IPv4 unicast, Description: This command displays the BGP routing table, which is crucial for analyzing route churn and identifying potential issues. The expected output will show the routing table structure and the presence of the default route (0.0.0.0/0).
```

```
Command: display ospf routing-table, Expected Output: Routing table for OSPF process, Description: This command displays the OSPF routing table, which helps in identifying adjacency loss and potential routing instability. The expected output will show the OSPF routing table structure and the presence of neighboring routers.

Command: display isis routing-table, Expected Output: Routing table for ISIS process, Description: This command displays the ISIS routing table, which aids in analyzing LSP flooding and potential routing issues. The expected output will show the ISIS routing table structure and the presence of neighboring routers.
```

Resolution Steps

Step 1: Step: Verify routing protocol configuration, Command: display bgp conf, Description: This command displays the BGP configuration, which helps in verifying that the configuration is correct and consistent. The expected output will show the BGP configuration parameters, such as AS number, router ID, and neighbor IP addresses.

Step 2: Step: Adjust routing protocol timers, Command: bgp timer timer-name time-value, Description: This command adjusts the routing protocol timers to reduce flapping and stabilize the routing table. The expected output will show the adjusted timer values.

Step 3: Step: Clear routing protocol cache, Command: clear bgp all, Description: This command clears the routing protocol cache, which helps in resolving routing table inconsistencies and reducing flapping. The expected output will show the cleared cache and the subsequent routing table rebuild.

Post-Check Verification

```
Command: display bgp routing-table ip-prefix 0.0.0.0/0, Expected Output: Routing table for VPN IPv4 unicast, Description: This command verifies that the routing table has stabilized and the default route (0.0.0.0/0) is present.

Command: display ospf routing-table, Expected Output: Routing table for OSPF process, Description: This command verifies that the OSPF routing table has stabilized and neighboring routers are present.

Command: display isis routing-table, Expected Output: Routing table for ISIS process, Description: This command verifies that the ISIS routing table has stabilized and neighboring routers are present.
```

Roll-back Procedures

- Step: Restore original routing protocol configuration, Command: undo bgp conf, Description: This command restores the original BGP configuration, which helps in reverting back to the previous stable state.

- Step: Revert adjusted routing protocol timers, Command: undo bgp timer timer-name time-value, Description: This command reverts the adjusted routing protocol timers to their original values, which helps in reverting back to the previous stable state.

- Step: Clear routing protocol cache (again), Command: clear bgp all, Description: This command clears the routing protocol cache again, which helps in resolving any further routing table inconsistencies and reducing flapping.